

Enterprise Digital Infrastructure – Lab Assignment

Spring 2026

1. Name Server performance.

- a. For a given Name Server, analyze whether and to what extent the query time depends on the type of query.
- b. For different Name Servers, analyze whether and to what extent the query time depends on their “distance”. For each Name Server, provide estimates of the RTT and of the number of hops.
- c. For a given Name Server and a given type of query, analyze whether and to what extent the query time depends on the DNS implementation (e.g., TCP, DoT, DoH).

2. HTTP/1.1 and HTTP/2 performance.

Clear the browser cache and the cookies of your browser and download the home page of the website of a University of your choice under HTTP/1.1 and HTTP/2. Make sure to choose a University that supports HTTP/2.

- a. Analyze the number and types of DNS queries, the number of TCP connections, the number of referenced objects, the number of web servers involved and the page load time. Does the website use cookies? How many?
- b. Reload the page and analyze its behavior with respect to the caching header fields specified in the response messages.
- c. Clear the browser cache and the cookies, modify the privacy policy of the browser with respect to cookies and reload the page. Does the number of cookies you receive change? Which types of cookies did you receive?
- d. Modify the maximum number of parallel TCP connections opened by the browser, clear the browser cache and reload the page. How many TCP connections does actually the browser use? Any change in the page load time?

3. Benchmarking.

- a. Test a website of your choice using **ab** and **h2load** benchmarking tools and assess whether and to what extent the various settings of the benchmark experiments (e.g., number of requests, concurrency level) affect the performance.